2.22 Equations of Motion of 4-vector Potential under GAUGE

$A_\mu \rightarrow A'_\mu = A_\mu - \partial_\mu \Lambda$ w/ $\Lambda(x^\mu) \equiv$ gauge f

a) Find the Eq. for which A'_μ satisfies the Lorentz gauge cond

Lorentz gauge condition $\equiv \partial_\nu A^\nu = 0$ (or $\partial^\nu A^\nu$

$$\partial^\mu A'_\mu = \xrightarrow{0}\partial^\mu A_\mu - \partial^\mu \partial_\mu \Lambda$$

$\Rightarrow$ the diff. eq. for Λ is: $\boxed{\Box^2 \Lambda = 0}$

or $\frac{1}{c^2}\frac{\partial^2 \Lambda}{\partial t} - \nabla^2 \Lambda = 0$

this is the wave equation; solution is $\Lambda = \lambda \exp$

$\Rightarrow$ can vary by amplitude (λ) and phase angle

b) Coulomb Gauge ($\nabla \cdot \vec{A} = 0$)

$$\nabla \cdot A'_\mu = \nabla \cdot A_\mu - \nabla \cdot (\partial_\mu \Lambda) = 0$$

$$0$$

$$\nabla \cdot \left(\frac{1}{c}\frac{\partial A}{\partial t} + \nabla \cdot \Lambda \right) = 0$$

$$\frac{1}{c}\frac{\partial(\nabla \cdot \Lambda)^{0}}{\partial t} + \nabla^2 \Lambda = 0 \quad \Rightarrow \boxed{\nabla^2 \Lambda = 0}$$

Λ is not unique